

Salil Lakra

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Summary

Software Engineer focused on distributed backend systems, AI-powered applications, and scalable full-stack platforms. Experienced with Go, TypeScript, Node.js, PostgreSQL, Redis, React, Docker, event-driven architecture, workflow orchestration, RAG, vector search, and asynchronous processing.

Education

Birla Institute of Technology, Mesra — B.Tech, Electronics and Communication Engineering (CGPA: 7.0/10) Aug 2024 – May 2028

Technologies

Languages: TypeScript, JavaScript, Go, Python, SQL, Java, C, C++ **Frontend:** React, Next.js, React Native, Tailwind CSS, Zustand, TanStack Query, React Flow

Backend: Node.js, Express.js, Hono, FastAPI, REST, GraphQL, JWT, OAuth, RBAC, WebSockets, SSE **AI/Search:** RAG, Gemini, Pinecone, pgvector, embeddings, semantic search, OCR, reranking

Data/Infra: PostgreSQL, MongoDB, MySQL, Redis, BullMQ, RabbitMQ, NATS JetStream, Docker, GitHub Actions, Linux, Nginx, Vercel, Cloudflare R2, Supabase

Experience

Full Stack Developer, Betiyan Nidhi Pvt. Ltd. — Ranchi Feb 2025 – May 2026

- Built cross-platform React Native applications supporting customer and field-agent operational workflows.
- Designed Node.js/Express backend services with MongoDB for secure financial and transactional workflows.
- Improved API performance through query optimization, indexing, and backend architecture improvements.
- Implemented JWT authentication, RBAC, and normalized data models for secure application access.

Projects

Forge – Distributed Workflow Orchestration Platform *Go, Next.js, PostgreSQL, NATS JetStream, Docker*

- Built a distributed workflow engine supporting DAG execution, cron scheduling, retries, asynchronous tasks, and durable workflow state.
- Designed event-driven execution with Go, NATS JetStream, and PostgreSQL; added React Flow workflow building and SSE-based execution monitoring.

github.com/salillakra/forge-workflow

Quill – AI-Powered Campus Operations & Assessment Platform *Next.js, Bun, Hono, PostgreSQL, Redis, BullMQ, Gemini, Pinecone*

- Built an AI campus platform for document intelligence, timetable management, assessment generation, student submissions, and AI-assisted grading across **5 roles**.
- Engineered asynchronous PDF/image processing with OCR, chunking, embeddings, extraction, generation, and grading using **BullMQ + Redis**.
- Implemented hybrid **RAG** using Pinecone + PostgreSQL lexical retrieval with **RRF** and reranking for grounded, page-cited responses.
- Added Cloudflare R2 object storage with presigned URLs and multipart uploads, plus Socket.IO pipeline-status updates and local fallback.

github.com/salillakra/quill

E-Summit 26 – Event Management Platform *Next.js, TypeScript, Supabase, Tailwind CSS*

- Built the official E-Summit platform serving **11K+ page views** with participant flows and a protected admin dashboard.
- Implemented Supabase authentication, OAuth, onboarding, protected routes, and role-based admin/moderator access.
- Developed event registration and team workflows with event-scoped membership validation and team-size enforcement.
- Built analytics for user growth, onboarding, registrations, teams, and engagement, plus real-time admin chat and presence.

github.com/EDC-BITM/e-summit26 esummit.edcbitmesra.in

Agile Turn Recruitment Suite – AI-Assisted ATS *Next.js, TypeScript, PostgreSQL, Redis, BullMQ, FastAPI, pgvector*

- Built an applicant-tracking platform covering jobs, candidates, applications, Kanban pipelines, interviews, reporting, and transactional email workflows.
- Developed a hybrid AI matching engine combining semantic similarity, skill overlap, experience, recency, resume quality, certifications, and location into explainable candidate-fit scores.
- Implemented recruiter natural-language search using embeddings, vector + full-text retrieval, RRF, hybrid ranking, caching, and role-scoped visibility.
- Built a FastAPI AI microservice for resume parsing and normalized embeddings with BullMQ workers for asynchronous resume parsing, embedding, email, and analytics jobs.

github.com/salillakra/Agile_Turn_Technology_LL

Achievements

- **3rd Prize, Internal Hackathon – Smart India Hackathon 2025, BIT Mesra:** Led a team developing a hardware-based Smart Agriculture solution for efficient cultivation in hilly regions.